Applied Statistics

307312 | University of Petra

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Start here

Work through the seven numbered **topic modules** below. The 8th-edition textbook supports the statistical topics, with reproducible R analyses in Quarto.

Start with Module 1: Introduction to R. Module 2 now begins with data types, measurement levels, and the descriptive statistics toolbox.

Study online or download

Student downloads provides the HTML website, a complete student RStudio project, and PDFs for offline reading, printing, or annotation.

Course modules

Module	Topic	Current material
1	Introduction to R	Four lessons and practice
2	Data and descriptive statistics	Lessons 1–4, practice, and integrated review available
3	Probability and distributions	Five lessons, practice, and integrated review available
4	Sampling and estimation	Four lessons, practice, and integrated review available
5	Hypothesis testing and group comparisons	Existing lesson; expansion planned
6	Regression	Planned
7	Forecasting and applied analysis	Planned

“Existing” means carried over from the previous course materials. “Planned” means a topic has been reserved in the course sequence, but its lessons have not yet been written.

How each module works

Each module has a clear starting page, lessons, and student exercises where available. Instructor solutions and course-planning notes are stored separately. Readings and data sources appear inside the relevant lesson.

Reference and supporting material

Our main statistical reference is Essentials of Modern Business Statistics with Microsoft Excel, **8th edition** (2020). We analyze data in **R** and write reproducible reports in **Quarto**.

Data and course resources provides dataset downloads and provenance guidance.